CLINICAL · LARYNX & AIRWAY

Cancer of the larynx

Laryngeal cancer is a smoker's disease with an unusual mercy: when it starts on the vocal fold it makes the voice hoarse while it is still small and has almost no lymphatics to spread through, so it is often found early and cured. Above or below the folds it is silent, spreads to the neck, and is found late. The rule is simple: hoarseness for more than three weeks in an adult, above all a smoker, means the larynx must be seen. Treatment then balances cure against voice, swallowing and the stoma.

AT A GLANCE

- Hoarse smoker gets a scope.
- Glottic tumours speak early; others spread early.
- Laser or radiotherapy for early disease.
- Chemoradiation or laryngectomy for advanced.
- Image before biopsy; NICE: offer T3 a choice of chemoradiotherapy or surgery; consider surgery first for T4a.

1 CHAPTER ONE Overview

Who and why.

1.1 Epidemiology

The second commonest cancer of the upper aerodigestive tract and about 1 to 5% of all malignancies; men are affected four to eight times as often as women, mostly in the sixth and seventh decades. In India it accounts for about 4% of male cancers. Glottic cancer is the commonest in most series (about 60%), supraglottic about 35 to 40%, and subglottic only 1 to 2%; supraglottic cancer predominates in some populations.

1.2 Causes

- **Tobacco:** the dominant cause; current smokers carry a ten- to twentyfold risk, which falls substantially after ten years of stopping.

- **Alcohol:** especially for supraglottic cancer; with tobacco the effects multiply (about fifteenfold).
- **Others:** previous radiation to the neck, papillomatosis (HPV 6 and 11, rarely malignant), occupational exposures (asbestos, nickel, mustard gas, wood dust, paints, diesel fumes), reflux, and genetic susceptibility.

1.3 Premalignant lesions

Hyperplasia, keratosis, mild, moderate and severe dysplasia and carcinoma in situ form a spectrum. Mild to moderate dysplasia progresses in about 7 to 11%, severe dysplasia in 18 to 30%; severe dysplasia and carcinoma in situ behave alike. Excisional biopsy (microflap or laser) is the diagnosis and usually the treatment; radiotherapy for multifocal disease or the frail; lifelong follow-up.

2 CHAPTER TWO Pathology and regions

Where it starts determines how it behaves.

2.1 Histology

Squamous cell carcinoma in 85 to 95%; glottic tumours tend to be well differentiated, supraglottic ones less so. Others: *verrucous carcinoma* (1 to 2%; exophytic, well-differentiated, with pushing margins; does not metastasise; treated by excision rather than radiation), spindle cell and basaloid carcinoma, neuroendocrine tumours, salivary carcinomas, chondrosarcoma (usually the cricoid) and lymphoma.

2.2 Regions

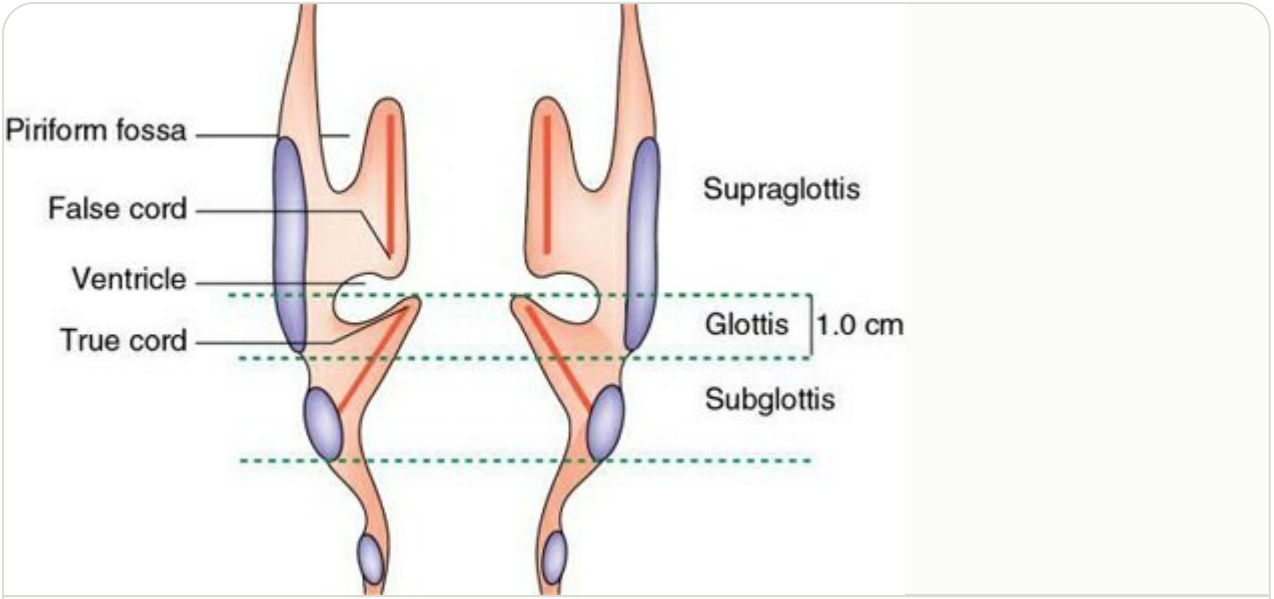


FIGURE 1 The regions of the larynx (AJCC): the glottis extends from the ventricle to 1 cm below the free edge of the fold; the subglottis from there to the lower border of the cricoid.

Dhingra, fig. 62.1

- **Supraglottis:** suprahyoid and infrahyoid epiglottis, aryepiglottic folds, arytenoids, false folds and ventricles. Embryologically from the third and fourth arches without midline fusion, with rich, bilateral lymphatics.
- **Glottis:** the true folds with the anterior and posterior commissures, extending 1 cm below the free edge. From the sixth arch, with scanty lymphatics.
- **Subglottis:** to the lower border of the cricoid.

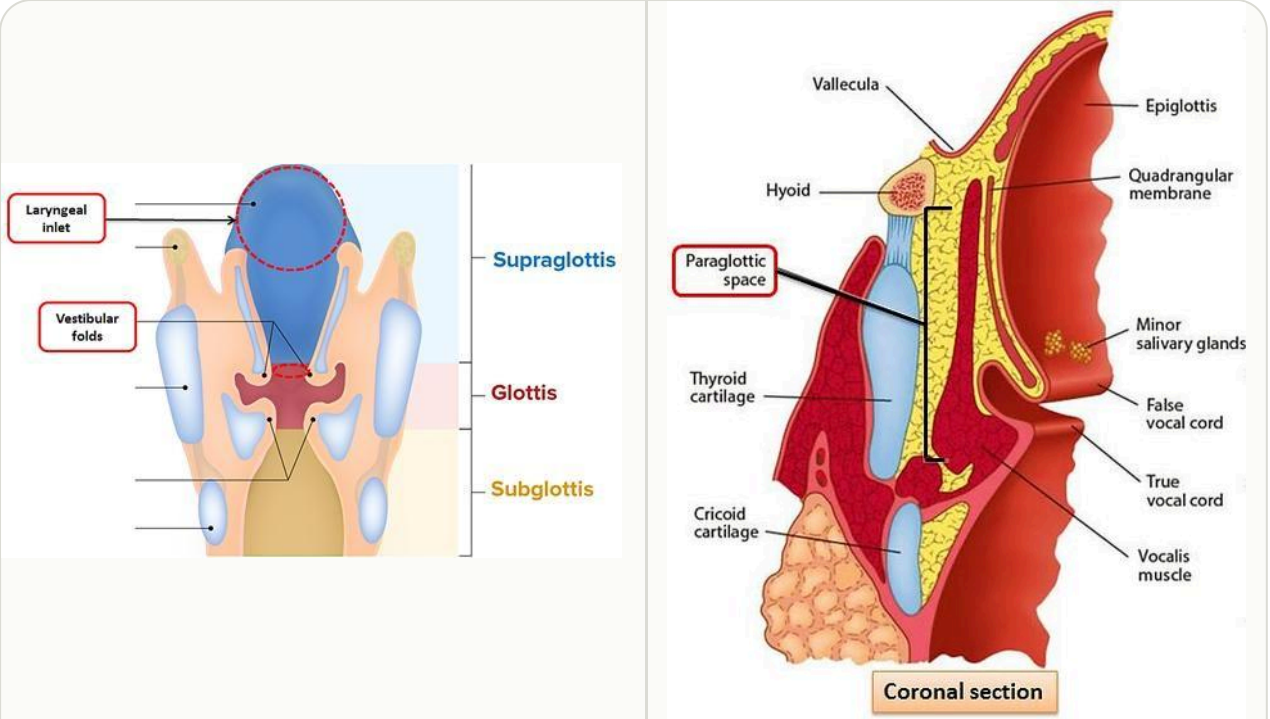


FIGURE 2 Supraglottis, glottis and subglottis, and the paraglottic space.
Anatomy atlas · ENT Steps p. 239, ENT Steps p. 244

2.3 Barriers and pathways

The conus elasticus and the quadrangular membrane resist spread; the *pre-epiglottic space* (reached through pits in the epiglottis) and the *paraglottic space* (lateral to the membranes) carry tumour silently across the ventricle (*transglottic* cancer) and up to the thyroid cartilage. At the anterior commissure, *Broyles' ligament* inserts into thyroid cartilage without perichondrium, a route into the cartilage.

3 CHAPTER THREE Supraglottic cancer

Silent and node-seeking.

3.1 Sites

Epiglottis, false folds and aryepiglottic folds, in that order. Suprahyoid epiglottic tumours tend to be exophytic, infrahyoid ones ulcerative.

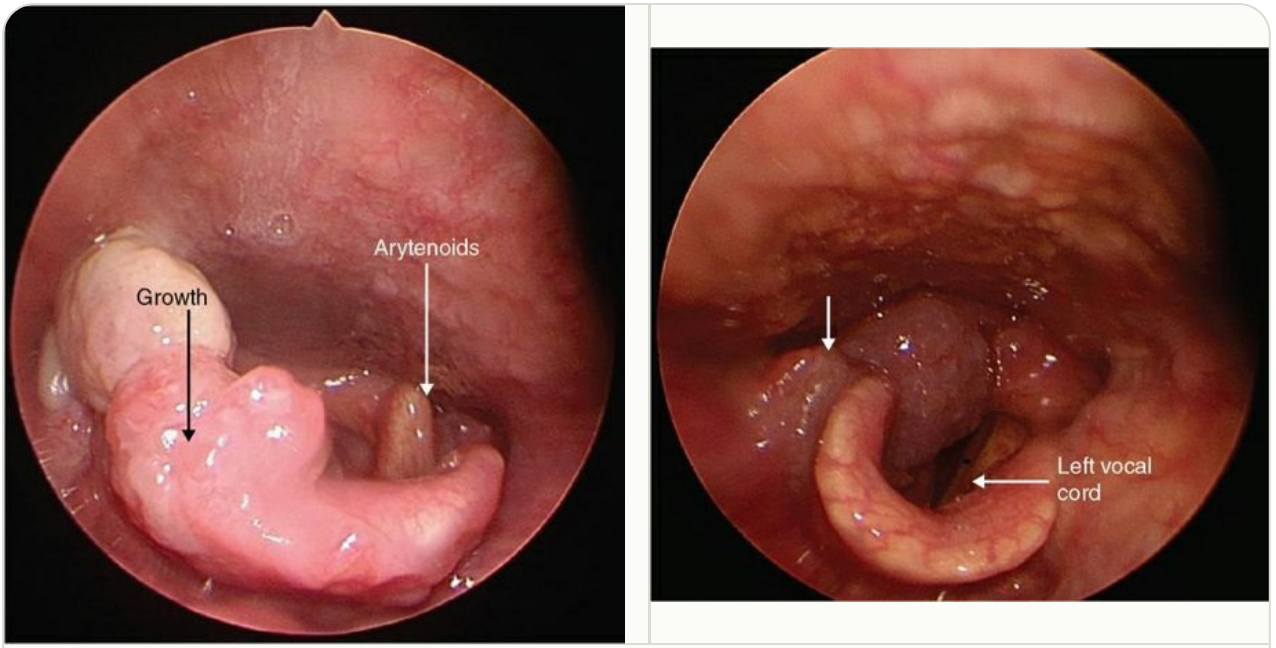


FIGURE 3 First, a supraglottic cancer of the epiglottis and right aryepiglottic fold. Second, a supraglottic cancer also invading the right piriform fossa (arrow).
Dhingra, figs 62.2 and 62.3

3.2 Spread

To the vallecula, base of tongue, piriform fossa and pre-epiglottic space, and through the thyroid cartilage. Nodes (levels II to IV) appear early: 25 to 50% at presentation, often bilateral, with occult nodes in 20 to 40% of N0 necks.

3.3 Symptoms

Often silent for a long time: discomfort or pain in the throat, a lump feeling, painful swallowing, referred otalgia, a neck mass; hoarseness, dyspnoea, halitosis and weight loss come late.

4 CHAPTER FOUR Glottic cancer

Loud and contained.

4.1 Site

The free edge and upper surface of the anterior and middle thirds of the fold.

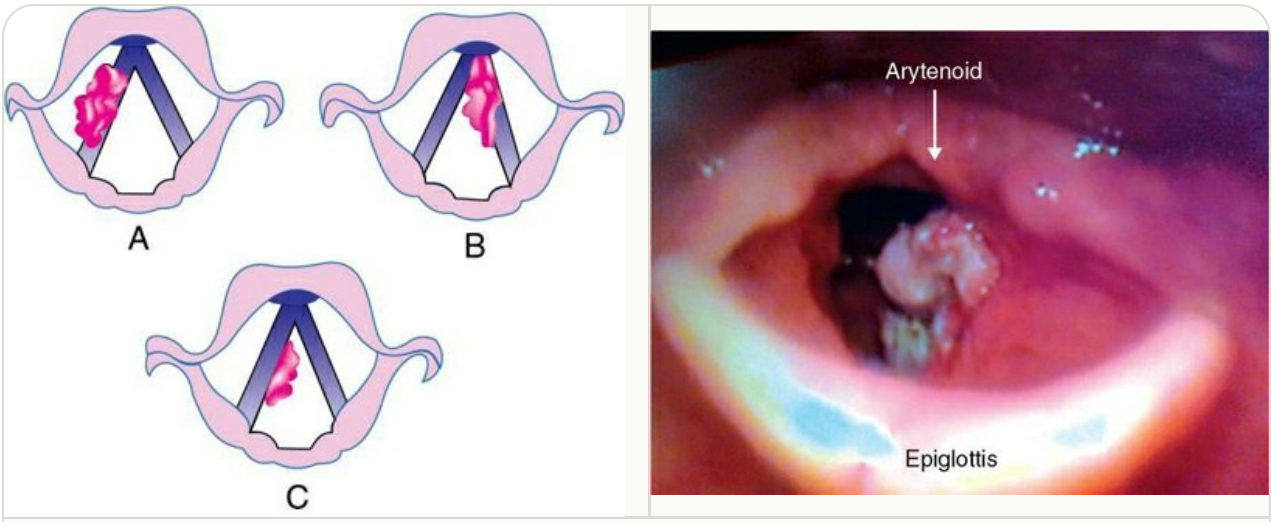


FIGURE 4 First, cancer of the larynx: (A) supraglottic, (B) glottic, (C) subglottic. Second, a cancer involving the supraglottic, glottic and subglottic areas on the left, on rigid endoscopy.
Dhingra, figs 62.4 and 62.5

4.2 Spread

Forwards to the anterior commissure and opposite fold; backwards to the vocal process and arytenoid; upwards to the ventricle and false fold; downwards to the subglottis; deeply into the thyroarytenoid muscle and paraglottic space. *Impaired mobility* means deep invasion of muscle, and *fixation* means invasion of the paraglottic space, cricoarytenoid joint or nerve: a bad prognostic sign. Because the folds have almost no lymphatics, node metastases are rare (under 5% in T1 and T2) until the tumour leaves the membranous fold.

4.3 Symptoms

Hoarseness early, because even a tiny lesion alters vibration; stridor when the tumour enlarges or fixes the fold.

CHAPTER FIVE

5 Subglottic cancer

Rare and late.

It starts on one side, spreads around the front and down into the trachea, through the cricothyroid membrane into the thyroid gland and strap muscles; nodes go to the prelaryngeal (Delphian), pretracheal, paratracheal (level VI), lower jugular and superior mediastinal groups. Stridor and dyspnoea are often the first symptoms; hoarseness is late, meaning spread to the undersurface of the fold, the thyroarytenoid or the recurrent nerve. It is usually advanced and poorly differentiated, and prone to stomal recurrence.

CHAPTER SIX

6 Diagnosis and staging

Seeing and mapping.

6.1 History and examination

Any patient of cancer age with hoarseness for more than two to three weeks needs laryngoscopy. Also ask about sore throat, otalgia, dysphagia, aspiration, dyspnoea, haemoptysis, weight loss, smoking and drinking, and lung function.

- **Laryngoscopy:** mirror, flexible or rigid video laryngoscopy shows the appearance (a nodule, ulcer, thickening or granular mass), mobility of each fold, and extent to the vallecula, base of tongue and piriform fossae. Narrow-band imaging helps separate malignant from benign lesions; office biopsy is possible with flexible scopes. *Stroboscopy* shows a stiff, non-vibrating segment in invasive cancer.
- **Neck:** extralaryngeal spread (a midline swelling over the cricothyroid membrane, tender cartilage from perichondritis, thyroid and strap muscle involvement, widening of the thyroid cartilage) and nodes (number, size, side, fixity).



FIGURE 5 Flexible fibreoptic laryngoscopy in the clinic.
Dhingra, fig. 62.6

6.2 Imaging

- **CT** of the neck with contrast: pre-epiglottic and paraglottic space invasion, cartilage destruction, extralaryngeal spread and nodes.
- **MRI:** more sensitive for cartilage invasion and for recurrence after radiotherapy.
- **CT of the chest:** lung metastases and second primaries (a solitary lung nodule is more likely a second primary).
- **PET-CT** for advanced disease and after chemoradiation (at 10 to 12 weeks) to detect residual disease.

6.3 Endoscopy under anaesthesia

Direct laryngoscopy shows the *hidden areas* (infrahyoid epiglottis, anterior commissure, ventricle, subglottis), maps the extent and allows biopsy; microlaryngoscopy for small fold lesions; panendoscopy for second primaries. Toluidine blue staining once guided biopsy of keratotic patches.

6.4 T staging (AJCC 8th edition)

T STAGING OF LARYNGEAL CANCER

CATEGORY	SUPRAGLOTTIS	GLOTTIS	SUBGLOTTIS
T1	One subsite, normal fold mobility	Folds only, normal mobility (T1a one fold, T1b both)	Limited to the subglottis
T2	More than one subsite of supraglottis, or glottis, or outside (base of tongue, vallecula, medial piriform wall), without fixation	Extends to supraglottis or subglottis, or impaired mobility	Extends to the folds, with normal or impaired mobility
T3	Fold fixation, or postcricoid area, pre-epiglottic space, paraglottic space, or inner cortex of thyroid cartilage	Fold fixation, paraglottic space, or inner cortex of thyroid cartilage	Fold fixation
T4a	Through the thyroid cartilage or beyond the larynx (trachea, soft tissues, deep tongue muscle, strap muscles, thyroid, oesophagus)	As supraglottis	Cricoid or thyroid cartilage, or beyond the larynx
T4b	Prevertebral space, encasing the carotid, or mediastinum	As supraglottis	As supraglottis

N staging follows the general head and neck rules, now including *extranodal extension*, which upstages nodes (N2a and N3b); Dhingra prints the earlier (6th edition) tables, based on size alone.

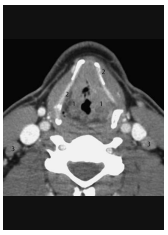
IMAGING IN THE APP 5 cases

Glottic and supraglottic tumours, pharyngeal tumours and synchronous malignancies on imaging.



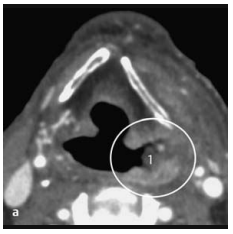
Glottic Tumor

Tumours · 5 images



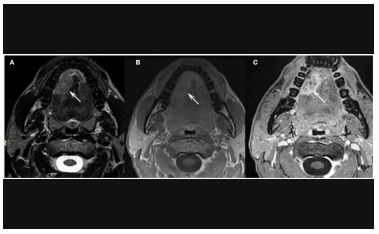
Supraglottic Tumor

Tumours · 9 images



Supraglottic and Pharyngeal Tumors

Tumours · 4 images



Squamous cell carcinoma: oral cavity, oropharynx and larynx

After surgery · 14 images



Multiple Synchronous Malignancies

Tumours · 3 images

CHAPTER SEVEN

7 Treatment

Cure, then function.

7.1 Principles

Early cancers (stage I and II) are treated by a single modality, radiotherapy or surgery, with similar control; the choice turns on voice, swallowing, access, fitness and preference. Advanced cancers (stage III and IV) need combined treatment, with organ preservation where it is safe and total laryngectomy where it is not.

7.2 Radiotherapy

For early tumours with mobile folds and no cartilage invasion or nodes: about 60 to 70 Gy over six weeks. T1 glottic cancer is controlled in about 90% or more with good voice; T2 in about 85% with normal mobility but much less (about 63 to 70%) when mobility is impaired. It spares the voice and avoids surgery but causes mucositis, oedema, dryness and occasional chondronecrosis, makes recurrence harder to detect, and cannot be repeated. Poor results with fixed folds, subglottic extension, cartilage invasion and nodes.

7.3 Transoral surgery

Transoral laser microsurgery (CO2 laser under the microscope), and more recently transoral robotic surgery, has largely replaced open partial laryngectomy for T1, T2 and selected T3 tumours: cordectomy for glottic cancer, supraglottic resection for supraglottic cancer. It is cheaper and quicker, avoids tracheostomy, keeps radiotherapy in reserve, and gives similar control to radiotherapy, but the voice may be poorer after wide cordectomy. It needs good exposure, and is contraindicated by fold fixation, interarytenoid involvement and poor lung function.

7.4 Open conservation surgery

Now used mainly in selected patients and after radiotherapy failure:

- **Cordectomy through a laryngofissure.**
- **Vertical partial (frontolateral, hemi-) laryngectomy:** one fold with the anterior commissure and adjacent thyroid cartilage.
- **Supraglottic (horizontal partial) laryngectomy:** removes the epiglottis, pre-epiglottic space, false folds and aryepiglottic folds above the ventricles; for T1 and T2 supraglottic cancer with mobile folds, clear of the anterior commissure, with good lung function, because every patient aspirates at first and must relearn swallowing.
- **Supracricoid partial laryngectomy:** removes both folds, the paraglottic spaces and the thyroid cartilage, rebuilt by cricohyoidopexy or cricohyoidoepiglottopexy; for selected T2 and T3 tumours with at least one mobile arytenoid.

7.5 Total laryngectomy

Removal of the whole larynx with the hyoid, pre-epiglottic space and upper trachea; the pharynx is closed and the trachea brought out as a permanent end stoma. Dhingra's indications:

- T3 and T4 tumours (fold fixed, cartilage invaded, extralaryngeal spread).
- Invasion of the thyroid or cricoid cartilage; both arytenoids; posterior commissure.
- Transglottic tumours with fold fixation.
- Failure after radiotherapy or conservation surgery.
- A non-functioning larynx with intractable aspiration.

Combined with neck dissection and postoperative radiotherapy as indicated; contraindicated by distant metastases.

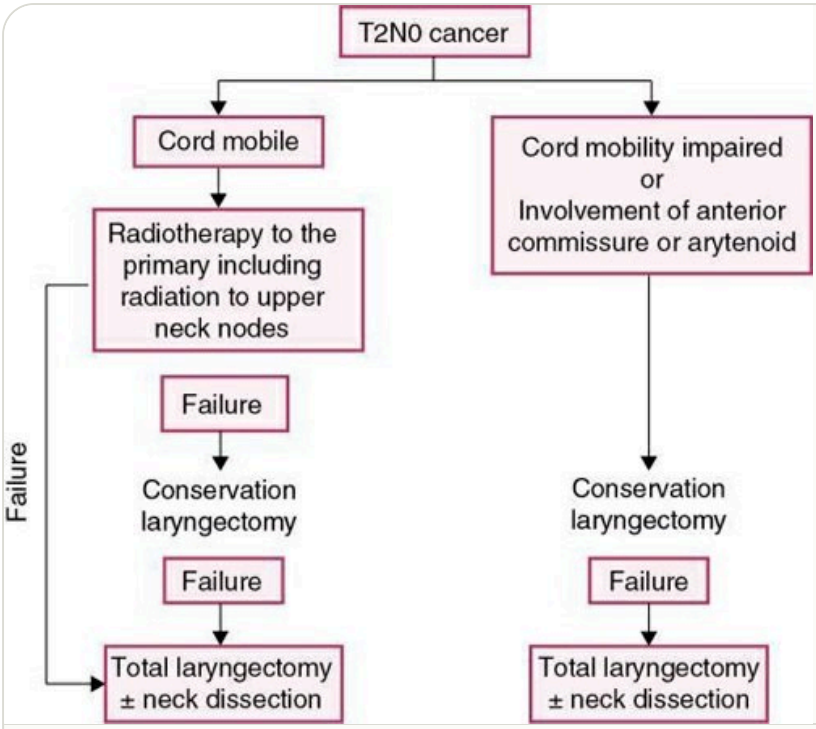


FIGURE 6 Treatment of T2N0 glottic cancer: radiotherapy when the fold is mobile and the commissure and arytenoid are spared; conservation laryngectomy when mobility is impaired or the commissure or arytenoid is involved; total laryngectomy for failure.
Dhingra, fig. 62.7

7.6 Organ preservation

The *VA Larynx trial* (1991) showed that induction chemotherapy (cisplatin and fluorouracil) followed by radiotherapy in responders gave the same survival as laryngectomy, with two-thirds keeping their larynx. *RTOG 91-11* (2003) showed that concurrent cisplatin with radiotherapy gave the best laryngeal preservation and locoregional control, although at ten years overall survival did not differ. Concurrent chemoradiation is now standard for most T3 and node-positive T2 cancers, with salvage laryngectomy for failure; cetuximab or carboplatin for those unfit for cisplatin. T4a tumours through cartilage, a non-functioning larynx and severe airway obstruction are better treated by laryngectomy and adjuvant therapy, which gives better control.

OPERATIVE GUIDES IN THEATRE 8 operations

Laser microsurgery, partial and total laryngectomy, laryngopharyngectomy and voice restoration.

Transoral CO2 Laser Microsurgery for Cancer of the Larynx
Laser
Horizontal Partial Laryngectomy
Horizontal partial
Vertical Partial Laryngectomy
Vertical partial
Supracricoid Partial Laryngectomy
Supracricoid
Supraglottic Transoral Robotic Surgery
TORS supraglottic
Total Laryngectomy
Total laryngectomy
Total Laryngopharyngectomy
Laryngopharyngectomy
Voice Restoration After Total Laryngectomy
Voice restoration

7.7 By site

- **Carcinoma in situ:** transoral laser excision (or stripping) with histology; radiotherapy if multifocal or invasive.
- **Glottic T1:** radiotherapy or laser cordectomy; with anterior commissure involvement, radiotherapy or a frontolateral laryngectomy.
- **Glottic T2:** radiotherapy if the fold is mobile and the commissure and arytenoid are free; otherwise conservation surgery; total laryngectomy if it recurs. The N0 neck is not treated (occult nodes under 5 to 25%).
- **Glottic T3 and T4:** chemoradiation for selected T3; total laryngectomy with neck dissection and postoperative radiotherapy for T4a.
- **Supraglottic T1 and T2:** radiotherapy, laser resection or supraglottic laryngectomy, always with *bilateral elective treatment of levels II to IV*.
- **Subglottic:** radiotherapy for early lesions; total laryngectomy with hemithyroidectomy or total thyroidectomy and level VI and paratracheal dissection, and radiotherapy including the superior mediastinum, for advanced lesions.

7.8 The neck

Clinically positive nodes need neck dissection or radiotherapy. After chemoradiation, a complete response on PET-CT at about twelve weeks allows

surveillance instead of planned neck dissection. Postoperative chemoradiation is given for positive margins or extranodal extension.

8 CHAPTER EIGHT After laryngectomy

Voice, smell and follow-up.

8.1 Voice restoration

COMMUNICATION AFTER TOTAL LARYNGECTOMY

METHOD	HOW IT WORKS	COMMENTS
Tracheo-oesophageal speech	A one-way valved prosthesis (Blom–Singer, Provox) in a puncture between trachea and oesophagus diverts lung air into the pharynx when the stoma is occluded	The best voice; most widely used; valve care and replacement needed
Oesophageal speech	Air is swallowed or injected into the upper oesophagus and released to vibrate the pharyngo-oesophageal segment	No device; rough but loud; hard to learn; six to ten words per breath
Electrolarynx	A battery-powered vibrator held against the neck (or a tube into the mouth) supplies sound that the mouth shapes	Easy to learn; mechanical voice
Writing and lip speech		Temporary

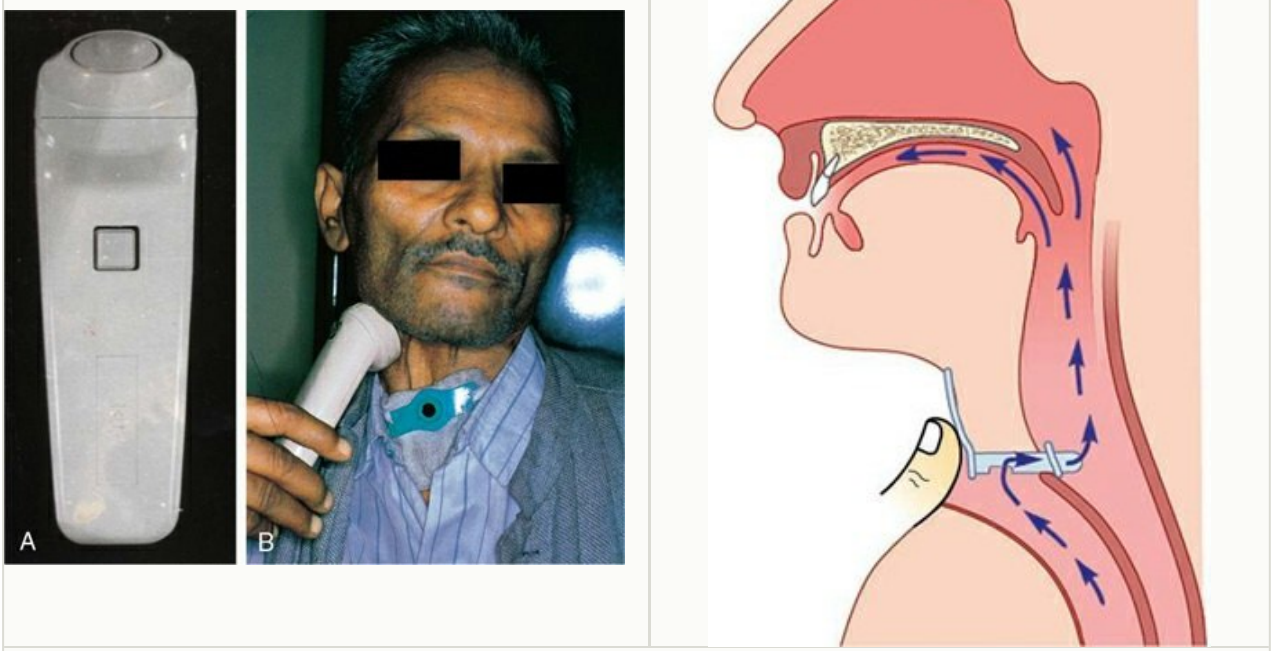


FIGURE 7 First, (A) an electrolarynx and (B) a laryngectomised patient using it. Second, tracheo-oesophageal speech: the thumb closes the stoma and air passes through the prosthesis into the oesophagus.

Dhingra, figs 62.8 and 62.9

8.2 Other rehabilitation

- **Smell:** lost because air no longer passes through the nose; the *polite yawning* manoeuvre (lowering the jaw and tongue with the lips closed) draws air through the nose.
- **Stoma care:** humidifying heat-moisture exchangers, protection from water.
- **Swallowing** and **thyroid function** (hypothyroidism in a fifth to two-thirds after treatment).

8.3 Complications

Pharyngocutaneous fistula (more after radiotherapy; reduced by a pectoralis major or free flap), stomal stenosis and stomal recurrence, pharyngeal stricture, chondronecrosis after radiotherapy, and a non-functioning larynx after chemoradiation.

8.4 Follow-up and recurrence

Every one to three months in the first year, less often thereafter, lifelong, looking for recurrence and second primaries. Recurrence shows as new oedema, pain, worse voice or reduced mobility; PET-CT and biopsy confirm it; about half are salvageable, usually by total laryngectomy (selected small recurrences by conservation surgery).

8.5 Prognosis

Five-year survival is about 90% for T1 glottic cancer, falling with stage; supraglottic and subglottic cancers do worse than glottic. Nodes, extranodal extension, cartilage invasion and subglottic extension are adverse.

NICE PATHWAY AND THE DECLINE IN SURVIVAL (OXFORD TEXTBOOK)

In the UK about 2,400 people a year develop laryngeal cancer, many from deprived communities; tobacco and alcohol account for about three-quarters of head and neck cancer. When cancer is suspected and the airway is safe, **cross-sectional imaging is done before biopsy**, because post-biopsy swelling can distort staging; ultrasound assesses the neck and allows node sampling, and panendoscopy looks for a synchronous primary (1 to 6%). Speech and language therapy assesses voice and swallow before treatment decisions.

Early disease (T1–T2, N0) is treated by transoral laser surgery or radiotherapy with similar control; overall survival ranges from about 45 to 77%. Laser is completed in a day and is cheaper, whereas radiotherapy takes three to seven weeks; attempts at a randomised trial have failed to recruit. Nodal spread is rare, so the neck is not routinely treated.

Advanced disease (T3–T4) is about 40% of cases. NICE advises offering T3 patients a choice of **chemoradiotherapy** or **surgery with adjuvant (chemo)radiotherapy**, and considering surgery first for T4a. The VA and RTOG 91-11 trials established larynx preservation with equivalent locoregional control, but population survival has fallen since practice shifted to non-surgical treatment, and patients who fail chemoradiotherapy face salvage laryngectomy with high complication rates and poor quality of life. Factors favouring primary laryngectomy are a poorly functioning larynx (aspiration, weight loss, tube feeding), a tracheostomy, T4 disease and patient preference. Quality of life is similar overall after surgical and non-surgical treatment, so

decisions should centre on the individual patient's values about voice and swallowing, made in the multidisciplinary team.

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Regions and subsites,** and why glottic cancer rarely spreads to nodes.
- **Pre-epiglottic and paraglottic spaces,** transglottic spread, Broyles' ligament.
- **Symptoms** by site; the hoarseness rule.
- **T staging** and the significance of fold mobility.
- **Treatment:** radiotherapy versus laser for early cancer; organ preservation (VA and RTOG 91-11); indications for total laryngectomy.
- **Voice rehabilitation** after laryngectomy.

9.2 Common pitfalls

COMMON PITFALLS

- Treating persistent hoarseness in a smoker with antibiotics and voice rest.
- Missing the anterior commissure, ventricle or subglottis without direct laryngoscopy.
- Forgetting bilateral neck treatment in supraglottic cancer.
- Offering organ preservation to a T4a tumour through cartilage.

Written for this app from Oxford Textbook of Otolaryngology ch. 55, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 62 and K. J. Lee ch. 44. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

LARYNX & TRACHEA · TUMOURS

Supraglottic Tumor

IMAGING FINDINGS

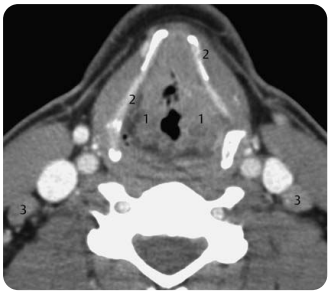
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

TOP DIFFERENTIALS

- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread



CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.

LARYNX & TRACHEA · TUMOURS

Supraglottic and Pharyngeal Tumors

- Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and overgrowth of the tumor on the contralateral side.
- Lymphomas (non-Hodgkin) initially might present as unilateral lymphadenopathy and ulcerative lesions.
- Evaluation of systemic disease and lymph nodes in other regions as well as secondary tumors in hypopharynx and larynx.
- Keep in mind that micrometastases may be radiologically occult.



CT with intravenous contrast enhancement, axial. Axial slice at the level of the thyroid cartilage shows a soft-tissue mass (outlined as…

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

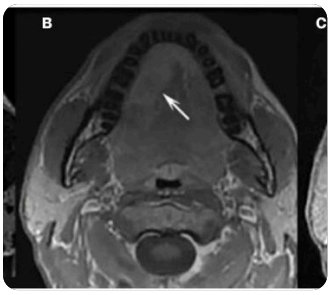
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



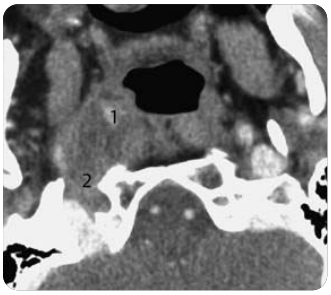
Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.

PHARYNX, TONSILS & OESOPHAGUS · TUMOURS

Multiple Synchronous Malignancies

- Staging is based on enlarged or pathologic-appearing lymph nodes, bilaterality, and spread of each tumor to the contralateral side.
- Lymphomas (non-Hodgkin) initially may present as unilateral lymphadenopathy and ulcerative lesions.
- Evaluation of systemic disease and lymph nodes in other regions.
- Keep in mind that (micro)metastases may be radiographically occult.



CT with intravenous contrast enhancement, axial.A lesion with malignant characteristics is seen in the right tonsillar fossa (1). The…